

Natural logs

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1 Understanding Natural Logs

1.1 At 100% Rate

\$1 in a bank account, 100% interest per year:

- Credited once, end of year: $1 + 1 \cdot 1 = 1 + 1 = \2
- Credited twice a year (50% each half-year), compounding: $(1 + 1 \cdot \frac{1}{2})^2 = \2.25
- Credited three times a year: $(1 + 1 \cdot \frac{1}{3})^3 \approx \2.37
- Credited four times a year: $(1 + 1 \cdot \frac{1}{4})^4 \approx \2.44

This suggests that for any compound interest at 100% for \$1, compounded n times, where x is the interest plus the starting amount:

$$x = \left(1 + 1 \cdot \frac{1}{n}\right)^n$$

- Plugging in 10: $(1 + 1 \cdot \frac{1}{10})^{10} \approx 2.594$
- 100: $(1 + 1 \cdot \frac{1}{100})^{100} \approx 2.705$
- Plugging in a arbitrary high number, say 1000000 into a calculator will get me close to the number we are converging on: $(1 + 1 \cdot \frac{1}{1000000})^{1000000} \approx 2.718$

1.2 Changing The Rate

For 200%:

$$\left(1 + \frac{2}{n}\right)^n$$

- At 1: $(1 + \frac{2}{1})^1 = 3$
- At 2: $(1 + \frac{2}{2})^2 = 4$
- At 10: $(1 + \frac{2}{10})^{10} \approx 6.192$
- At 100: $(1 + \frac{2}{100})^{100} \approx 7.245$ At 1000000: $(1 + \frac{2}{1000000})^{1000000} = 7.388$

It is apparent that $2.718 \cdot 2 \approx 5.4 - 5.5$, so that's not the relationship. It looks to be fairly close to the square, 2.718^2 should be somewhere between 4 and 9. Into the calculator, and $2.718^2 \approx 7.388$. This indicates:

$$x = \left(1 + \frac{r}{n}\right)^n$$

Where x is the starting \$1 + compound interest, at rate r compounded n times. If we let n approach infinity where $r = 1$, we know from above that:

$$\lim_{n \rightarrow \infty} = e \approx 2.718$$

So if $e^2 \approx 7.388$ and the above $(1 + \frac{2}{n})^n \approx 7.388$, when $\lim_{n \rightarrow \infty}$, it indicates:

$$\left(1 + \frac{r}{n}\right)^n = e^r$$

To test, we'll try $\frac{1}{2}$: $(1 + \frac{\frac{1}{2}}{1000000})^{1000000}$

Knowing $e^{\frac{1}{2}} = \sqrt{e}$, $\sqrt{2} \approx 1.41$ and $\sqrt{3} \approx 1.73$ (had to use calculator for that), I can imagine $\sqrt{e} \approx \frac{e}{2}$ (Mistake, 1.141 is not less than half of 2)

Using a calculator, $\sqrt{e} \approx 1.649$

$1.649 \cdot 2 = 3.298$ so obviously not that case.

Still, to verify e^r our equation should converge on 1.649, which it does

1.3 Questions

1. Evaluate three, using the "raised to what" reading: $\ln(e^3)$, $\ln(1)$, $\ln(\sqrt{e})$
2. Solve for x : $\ln(x) = 2$
3. Solve and explain in one sentence: $e^x = 1$
4. Application and real world one: \$100 at 5% annual interest for 10 years, compounded continuously. What's the balance? (Use your law - the answer is $100 \cdot e^{rt}$; work out what r and t contribute.)
5. Why, no computation: why is $\ln(1) = 0$ something you know without a calculator?

1.4 Answers

1. Let $\ln(e^3) = x$ and $\ln(\sqrt{e}) = y$.

$$\begin{aligned}\ln(e^3) &= x \\ e^x &= e^3\end{aligned}$$

Since e^n is one-to-one for any n :

$$x = 3$$

For any base b including $\ln$ $\log_b(1) = 0$, as $b^0 = 1$

$$\begin{aligned}y &= \ln(\sqrt{e}) \\ e^y &= \sqrt{e} \\ \sqrt{e} &= e^{\frac{1}{2}} \\ y &= \frac{1}{2}\end{aligned}$$

2.

$$\begin{aligned}\ln(x) &= 2 \\ x &= e^2 \\ x &\approx 7.388\end{aligned}$$

3. In $e^x = 1$, $x = 0$. This is true for number n : $n^0 = 1$

4. x will be our compound interest + original sum (final balance). i will be our initial sum. n will be our compounding number, which will increase to infinity.

$$x = i \times \left(1 + \frac{r \times t}{n}\right)^n$$

We know that r goes into the numerator in the function; t (time) goes in there too as we are calculating 5% a year over 10 years, so 50%. In a noncontinuous function, t would also multiply with n , as 12 compounds a year over 10 years would be 120 compounds. $r \cdot t$ is the total interest rate, so:

$$\begin{aligned}x &= i \cdot e^{rt} \\ &= 100 \cdot e^{0.05 \cdot 10} \\ x &= \$164.87\end{aligned}$$

5. Any $\log(1) = 0$ including $\ln$ because base (b) $b^0 = 1$.